Notes: Snowberg, Wolfers, and Zitzewitz (QJE, 2007)

Partisan Impacts on the Economy: Evidence from Prediction Markets and Close Elections

1 Big picture

This note summarizes the paper, which uses prediction-market probabilities, high-frequency financial futures, and close-election events to estimate how U.S. presidential election outcomes causally affect financial-market expectations about equities, bond yields, oil prices, and exchange rates.

Core question. Do presidential election outcomes have economically meaningful effects on financial markets and, by implication, expected macroeconomic policy and activity?

Main identification problem. A simple regression of asset prices on election probabilities is not causal because economic news affects both financial markets and the probability that a party wins. For example, good economic news may raise equity prices and also improve the incumbent party's election odds.

Main solution. The authors exploit sudden, plausibly exogenous shifts in beliefs about the election winner during election night. In 2004, flawed exit-poll information sharply reduced the predicted probability of a Bush victory, and the subsequent vote count sharply reversed that belief. In 2000, the Florida calls and rescissions generated similar election-night belief shocks.

Central finding. Financial markets behaved as if a Republican/Bush presidency implied higher equity prices, higher nominal and real bond yields, higher short-run oil prices, and a stronger dollar than a Democratic/Kerry or Gore presidency. In historical data since 1880, surprise Republican victories are associated with a roughly 2–3 percent increase in equity values.

2 Incremental contribution of the paper

- **Identification contribution.** The paper shifts from low-frequency correlations between political expectations and asset prices to high-frequency election-night shocks, reducing reverse-causality and omitted-variable concerns.
- **Measurement contribution.** It combines prediction-market probabilities with intraday financial futures, allowing the coefficient on election probabilities to be interpreted as the financial-market effect of a change from one candidate to another.
- **Historical contribution.** It reconstructs political prediction-market probabilities for presidential elections back to 1880 and uses the magnitude of electoral surprises rather than only the winner's party.
- **Political-economy contribution.** The evidence challenges strong policy-convergence views: markets price different economic outcomes under different parties.
- **Methodological contribution.** The paper illustrates how prediction markets can serve as real-time instruments for changing beliefs in event studies.

3 Data and measurement

Prediction-market data. For the 2004 election, the key political variable is the price of a Tradesports contract paying $10 if George W. Bush is re-elected. The price is interpreted as the market-implied probability of a Bush victory. Observations are collected every ten minutes from noon on November 2, 2004 to 6 a.m. on November 3.

Financial data. The paper matches prediction-market prices to high-frequency futures prices: S&P 500, Dow Jones, Nasdaq 100, currency futures, Treasury futures, and crude oil futures. Futures are used because they trade outside regular market hours.

2000 election data. There was no suitable prediction-market contract tracking Electoral College victory in 2000, so the authors use election-night news events and pre-election bookmaker odds to bound the effect of Bush versus Gore.

Historical data. For elections from 1880 to 2004, the authors combine historical betting markets, Curb Market data, bookmakers, the Iowa Electronic Markets, and Tradesports. The key historical variable is the partisan shock:

$$\text{Partisan Shock}_t = \mathbf{1}\{\text{Republican president elected}_t\} - P_{t-1}(\text{Republican president}).$$

4 Empirical design

4.1 High-frequency election-night regression

The main estimating equation is

$$\Delta \log(\textit{Financial variable}_t) = \beta\, \Delta P_t(\text{Bush victory}) + \varepsilon_t.$$

The coefficient β measures the percentage difference in the financial variable under a Bush presidency relative to a Kerry presidency. The identifying assumption is that, over ten- or thirty-minute election-night windows, changes in the prediction-market probability are driven by election news rather than by unrelated financial news.

4.2 Why high frequency matters

The same relation estimated with daily pre-election data is badly confounded. For example, rising equity prices before the election may reflect a stronger economy that helps Bush's re-election probability, not a causal effect of Bush on equities. Election-night shocks are cleaner because they move political beliefs sharply while leaving little time for unrelated macroeconomic news.

4.3 Historical event-study specification

For earlier elections, the paper uses a narrower-than-usual event window: stock returns from the pre-election close to the post-election close. The key explanatory variable is the prediction-market surprise about whether a Republican president is elected. The historical design is noisier than the 2004 intraday design, but it enables a long-run comparison across many election cycles.

5 Main results

5.1 2004 Bush versus Kerry

The election-night estimates imply that a Bush presidency, compared with a Kerry presidency, raised major equity index futures by about 1.5–2.4 percent. The effect appears across the S&P 500, Dow Jones Industrial Average, and Nasdaq 100.

Bond yields also rise under Bush. The 2004 high-frequency estimates suggest increases of roughly 10–12 basis points in two-year and ten-year Treasury yields. The comparison with inflation-indexed securities indicates that this movement is mostly in real rates, not expected inflation.

Oil prices rise in the near term under Bush, while the dollar tends to strengthen. The authors interpret these movements as consistent with markets expecting more expansionary fiscal policy and possibly stronger demand under Bush, though the evidence does not identify exactly which policy channel drives the response.

5.2 Time-series bias

Daily pre-election correlations imply much larger equity effects and sometimes the opposite sign for oil prices. This contrast shows why conventional time-series designs are misleading: they mix the effect of politics on markets with the effect of economic conditions on electoral expectations.

5.3 2000 Bush versus Gore

The 2000 Florida-related news shocks produce bounded effects that are consistent with the 2004 results: a Bush victory is associated with higher equity prices and a stronger dollar. This helps distinguish a partisan effect from a generic effect of re-electing an incumbent.

5.4 Historical evidence since 1880

Using historical prediction-market surprises, the authors find that equity markets generally rise after surprise Republican victories and fall after surprise Democratic victories. The estimated effect of electing a Republican president is about 2–3 percent for value-weighted equity returns.

For bond yields, the historical evidence is different before and after Reagan. Before 1980, bond markets show little systematic partisan response. Since 1980, Republican victories are associated with higher long-term bond yields, consistent with a post-1980 change in expectations about deficits, debt, and fiscal policy.

6 Economic interpretation

Partisan effects are priced. The evidence suggests that financial markets expected materially different economic policies under Republican and Democratic presidents.

Equity effects are not welfare effects. Higher equity prices under Republicans could reflect policies favorable to capital, incumbent firms, current shareholders, or near-term activity. This does not imply higher aggregate welfare.

Bond-yield effects point to fiscal expectations. The rise in real yields under Bush is difficult to reconcile with a simple view that Republican presidents are always expected to be more fiscally conservative. Instead, it is consistent with expectations of more expansionary fiscal policy or higher real rates.

Prediction markets improve event studies. A central methodological insight is that the size of the election surprise matters. Treating all Republican victories equally discards valuable information about how much news the election outcome reveals.

7 Limitations

- The main 2004 evidence identifies financial-market expectations about Bush versus Kerry, not all possible Republican versus Democratic candidates.
- The paper measures effects on asset prices and yields, not direct effects on GDP, welfare, or household outcomes.
- Historical event windows are noisier because many non-election shocks may occur between the pre-election and post-election market close.
- Prediction-market prices reflect beliefs of market participants and may differ from representative-voter expectations.
- The paper cannot fully decompose whether equity responses arise from expected taxes, regulation, fiscal policy, labor-market policy, entry policy, or aggregate demand.

8 Tables and figures

8.1 Figures I and II: Election-night movements in 2004 and 2000

Read: Figure I plots the 2004 Tradesports probability of a Bush victory against the S&P 500 future. Figure II shows analogous financial-market movements during the 2000 Florida vote-count sequence.

Economic message: Financial prices move tightly with sudden changes in expected election outcomes. This visual evidence motivates the paper's high-frequency regression design.

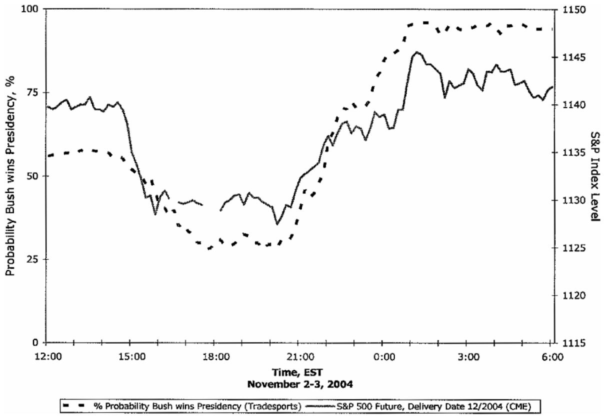
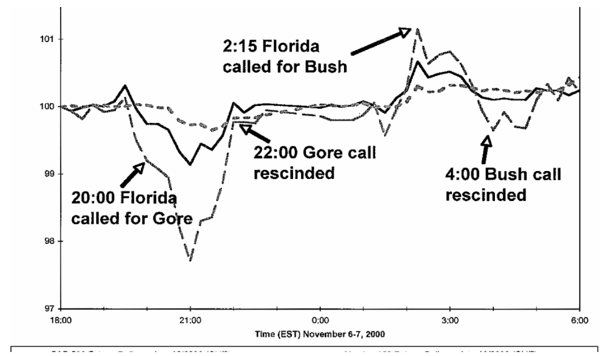


FIGURE I
The S&P 500 is Higher under a Bush versus Kerry Presidency

(a) Figure I: S&P 500 and Bush re-election probability in 2004



(b) Figure II: Election-night financial movements in 2000

8.2 Tables I and II: High-frequency effects and bond-yield decomposition

Read: Table I estimates the effect of Bush versus Kerry on equity indices, the dollar, oil futures, and Treasury yields. Table II uses inflation-indexed and nominal bond assets to distinguish real-rate movements from inflation-expectation movements.

Economic message: Bush's re-election was associated with higher equity valuations and higher bond yields. The bond response appears mainly to reflect real-rate movements rather than inflation expectations.

TABLE I

EFFECTS OF BUSH VERSUS KERRY ON FINANCIAL VARIABLES

Dependent variable	10-minute first differences: Estimated effect of Bush presidency	n	30-minute first differences: Estimated effect of Bush presidency	n
Dependent variable: ΔLog(Financial Index)				
S&P 500	0.015***	104	0.021***	35
	(0.004)		(0.005)	
Dow Jones industrial average	0.014***	84	0.021***	29
	(0.005)		(0.006)	
Nasdaq 100	0.017***	104	0.024***	35
	(0.006)		(0.008)	
U.S. dollar (vs. trade-weighted basket)	0.004	93	0.005**	34
	(0.003)		(0.003)	
Dependent variable: ΔPrice				
Light crude oil futures				
December '04	1.110***	88	1.706**	29
	(0.371)		(0.659)	
December '05	0.652*	85	1.020	28
	(0.375)		(0.610)	
December '06	−0.580	63	−0.666	21
	(0.783)		(0.863)	
Dependent variable: ΔYield				
2-Year T-note future	0.104*	84	0.108***	30
	(0.058)		(0.036)	
10-Year T-note future	0.112**	91	0.120**	31
	(0.050)		(0.046)	

White 1980 standard errors in parentheses. The sample period is noon EST on 11/2/2004 to 6 A.M. on 11/3/2004. Election probabilities are the most recent transaction prices collected every ten minutes from Tradesports.com, S&P, Nasdaq, and foreign exchange futures are from the Chicago Mercantile Exchange; Dow and bond futures are from the Chicago Board of Trade, while oil futures data are from the New York Mercantile Exchange. Equity, bond, and currency futures have December 2004 delivery dates. Yields are calculated for the Treasury futures using the daily yields reported by the Federal Reserve for 2- and 10-year Treasuries and projecting forward and backward from the bond market close at 3 P.M. using future price changes and the future's durations of 1.96 and 7.97 reported by CBOT. The trade-weighted currency portfolio includes six currencies is the CME traded futures (the Euro, Yen, Pound, Australian and Canadian dollars

(a) Table I: Effects of Bush versus Kerry on financial variables

TABLE II

CHANGES IN BOND YIELDS WERE UNRELATED TO CHANGES IN INFLATION EXPECTATIONS

	1st natural experiment 3–4 P.M. 11/2/2004 ΔProb(Bush) = −12%		2nd natural experiment 4 P.M.–9:30 A.M. 11/2/2004–11/3/2004 ΔProb(Bush) = 55%	
	Δ(Yield)	Wald estimator: $\frac{\Delta(\text{Yield})}{\Delta\text{Prob(Bush)}}$	Δ(Yield)	Wald estimator: $\frac{\Delta(\text{Yield})}{\Delta\text{Prob(Bush)}}$
Inflation-indexed yields (%)				
Lehman TIPS ETF ("TIP")	−0.020	0.16	0.040	0.11
Non-index yields (%)				
Lehman 7–10-year Treasury ETF ("IEF")	−0.020	0.16	0.061	0.15
CBOT 10-year Treasury Note	−0.009	0.07	0.050	0.11
CBOT 2-year Treasury	−0.015	0.13	0.030	0.08

For the TIP and IEF exchange traded funds (ETF), the implied yield for 3 P.M. is taken to be the constant-maturity daily yield calculated by the Federal Reserve for TIPS and Treasuries with the closest maturity to average holdings of the ETFs (seven years in both cases). For the 10- and 2-year CBOT Treasury futures, the 10 and 2-year series are used. These yields are then projected forward using price changes and the average duration of the funds holdings, as reported by Morningstar in December 2004.

(b) Table II: Bond yields and inflation expectations

8.3 Tables III and IV: Pre-election bias and the 2000 comparison

Read: Table III shows that daily pre-election correlations between Bush's re-election probability and financial variables produce very different estimates from the election-night design. Table IV bounds the effect of Bush versus Gore in 2000.

Economic message: Low-frequency correlations are contaminated by reverse causality. The 2000 election provides a second event-style comparison that is broadly consistent with the 2004 partisan interpretation.

TABLE III

Re-election Probabilities and Financial Variables Through the Campaign

Independent Variable: ΔBush election probability	Daily differences Estimate	n	5-day differences Estimate	n	20-day differences Estimate	n
Dependent Variable: ΔLog(Financial Index)						
S&P 500	0.087** (0.034)	321	0.128** (0.062)	317	0.243** (0.065)	302
Dow Jones industrial average	0.093*** (0.032)	321	0.145** (0.061)	317	0.275*** (0.090)	302
Nasdaq 100	0.143** (0.062)	321	0.212** (0.098)	317	0.299*** (0.108)	302
U.S. Dollar (vs. trade-weighted basket)	0.040** (0.019)	321	0.017 (0.022)	317	−0.022 (0.047)	302
Dependent Variable: ΔPrice						
Light crude oil futures (near month)	0.390 (4.504)	318	−7.221 (7.188)	314	12.547* (6.793)	299
Dependent Variable: ΔYield						
10-Year T-bill yield	1.130*** (0.373)	321	0.463 (0.489)	317	−0.028 (0.718)	302

Newey-West [1987] standard errors in parentheses, allowing for autocorrelation over 1, 5, and 20 lags, respectively. Financial variables are daily closing prices. The U.S. dollar is measures relative to a trade-weighted basket of the same currencies as in Table I. Sample covers all trading days from June 2003 to October 2004.

(a) Table III: Re-election probabilities and financial variables during the campaign

TABLE IV

Natural Experiments in 2000 Provide Estimates in Line with Those from 2004

	1st natural experiment 6 P.M.–9 P.M. 11/6/2000 0percent ≤ ΔProb(Bush) ≤ 60%		2nd natural experiment 9 P.M.–2:15 A.M. 11/6/2000–11/7/2000 0% ≥ ΔProb(Bush) ≥ 100%	
	%Δ(Price)	Wald estimator: %Δ(Price) / ΔProb(Bush)	%Δ(Price)	Wald estimator: %Δ(Price) / ΔProb(Bush)
S&P 500 (%)	−0.9	≥1.4	1.6	≥1.6
Nasdaq 100 (%)	−2.3	≥3.8	3.5	≥3.5
U.S. dollar (vs. Trade-weighted basket) (%)	−0.3	≥0.5	0.6	≥0.6

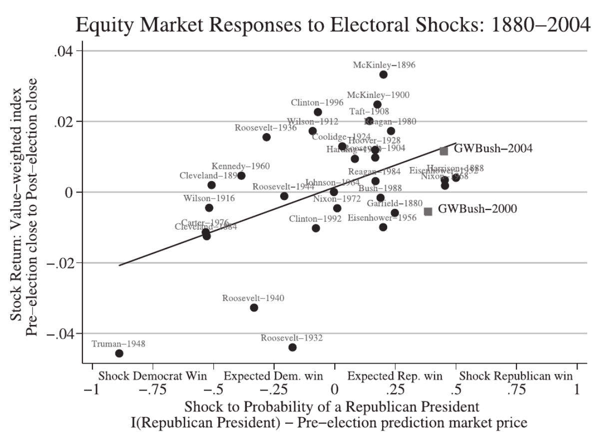
Election night events indicate that the highest probability of a Bush loss occurred around 9 P.M. and the

(b) Table IV: Natural experiments in the 2000 election

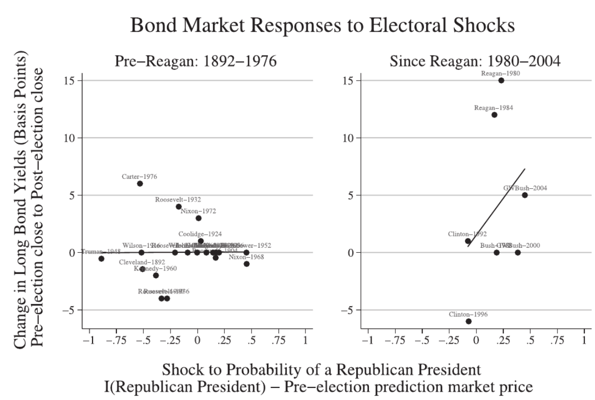
8.4 Figures III and IV: Historical partisan shocks

Read: Figure III plots stock returns around presidential elections against the prediction-market shock to the probability of a Republican president. Figure IV repeats the logic for bond yields, separating the pre-Reagan and post-Reagan eras.

Economic message: Equity markets historically respond positively to surprise Republican victories. Bond markets show little partisan response before 1980, but a positive Republican-yield response after Reagan.



(a) Figure III: Equity-market responses to electoral shocks



(b) Figure IV: Bond-market responses to electoral shocks

8.5 Table V: Historical equity-return regressions

Read: Table V compares a simple Republican-winner indicator with a prediction-market-based partisan shock. Using the magnitude of the surprise produces statistically stronger and more interpretable estimates.

Economic message: The historical average effect of a Republican presidential victory is roughly a 2–3 percent increase in equity returns, and this result is driven by partisan surprise rather than incumbency.

TABLE V

THE EFFECT OF A REPUBLICAN ON VALUE-WEIGHTED EQUITY RETURNS

	Dependent Variable: Stock returns from election-eve close to post-election close			
I (GOP President)				
[As in Santa-Clara and	0.0129			
Valkanov]	(0.0089)			
ΔProb(GOP President)		0.0297**	0.0249***	0.0242***
(Prediction markets)		(0.118)	(0.0081)	(0.0084)
ΔProb(Incumbent party				
elected)				−0.0038
(Prediction markets)				(0.0085)
	−0.0102	−0.0027	0.0014	0.0013
Constant	(0.0059)	(0.0040)	(0.0028)	(0.0028)
Sample	1928–1996	1928–1996	1880–2004	1880–2004
N	18	18	32	32

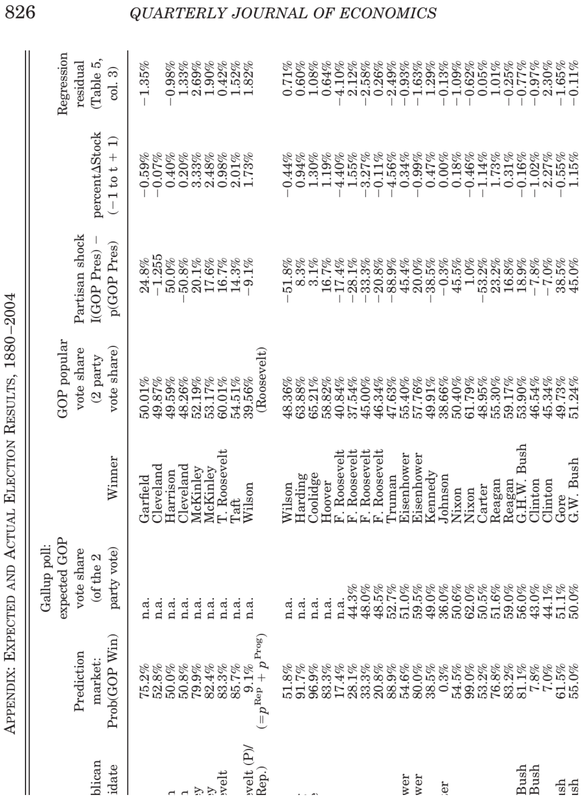
White standard errors in parentheses. See Appendix for further details on construction of variables.
***, **, and * denote statistically significant at 1 percent, 5 percent, and 10 percent

Table V: The effect of a Republican on value-weighted equity returns

8.6 Appendix table: Expected and actual election results, 1880–2004

Read: The appendix records the prediction-market probabilities, Gallup polling information, election winners, popular vote shares, partisan shocks, and stock returns used to build the long historical sample.

Economic message: The paper's historical analysis depends on measuring the surprise component of each election outcome, not merely whether a Republican or Democrat won.



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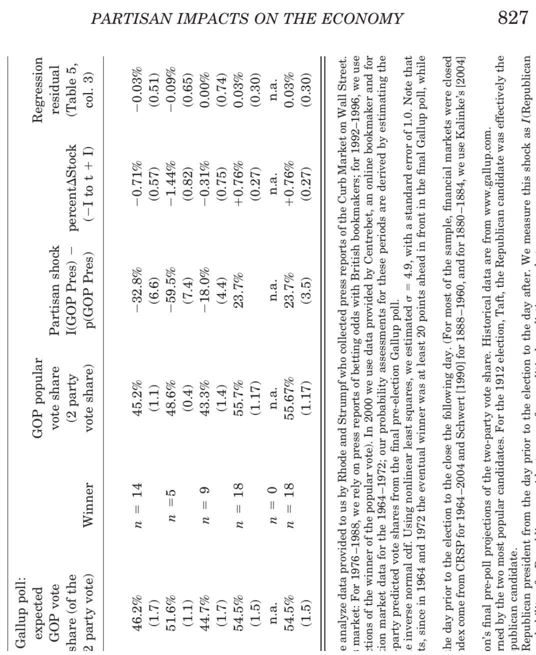
APPENDIX: EXPECTED AND ACTUAL ELECTION RESULTS, 1880–2004

…blican …idate	Prediction market: Prob(GOP Win)	Gallup poll: expected GOP vote share (of the 2 party vote)	Winner	GOP popular vote share (2 party vote share)	Partisan shock I(GOP Pres) − p(GOP Pres)	percentΔStock (−1 to t + 1)	Regression residual (Table 5, col. 3)
	75.2%	n.a.	Garfield	50.01%	24.8%	−0.59%	−1.35%
	52.8%	n.a.	Cleveland	49.87%	−1.255	−0.07%	
…n	50.0%	n.a.	Harrison	49.59%	50.0%	0.40%	−0.98%
…n	50.8%	n.a.	Cleveland	48.26%	−50.8%	0.20%	1.33%
…y	79.9%	n.a.	McKinley	52.19%	20.1%	3.33%	2.69%
…y	82.4%	n.a.	McKinley	53.17%	17.6%	2.48%	1.90%
…velt	83.3%	n.a.	T. Roosevelt	60.01%	16.7%	0.98%	0.42%
	85.7%	n.a.	Taft	54.51%	14.3%	2.01%	1.52%
…velt (P)/ Rep.)	9.1% ($=p^{Rep}+p^{Prog}$)	n.a.	Wilson	39.56% (Roosevelt)	−9.1%	1.73%	1.82%
	51.8%	n.a.	Wilson	48.36%	−51.8%	−0.44%	0.71%
	91.7%	n.a.	Harding	63.88%	8.3%	0.94%	0.60%
	96.9%	n.a.	Coolidge	65.21%	3.1%	1.30%	1.08%
	83.3%	n.a.	Hoover	58.82%	16.7%	1.19%	0.64%
	17.4%	n.a.	F. Roosevelt	40.84%	−17.4%	−4.40%	−4.10%
	28.1%	44.3%	F. Roosevelt	37.54%	−28.1%	1.55%	2.12%
	33.3%	48.0%	F. Roosevelt	45.00%	−33.3%	−3.27%	−2.58%
	20.8%	48.5%	F. Roosevelt	46.34%	−20.8%	−0.11%	0.26%
	88.9%	52.7%	Truman	47.63%	−88.9%	−4.56%	−2.49%
…wer	54.6%	51.0%	Eisenhower	55.40%	45.4%	0.34%	−0.93%
…wer	80.0%	59.5%	Eisenhower	57.76%	20.0%	−0.99%	−1.63%
	38.5%	49.0%	Kennedy	49.91%	−38.5%	0.47%	1.29%
…er	0.3%	36.0%	Johnson	38.66%	−0.3%	0.00%	−0.13%
	54.5%	50.6%	Nixon	50.40%	45.5%	0.18%	−1.09%
	99.0%	62.0%	Nixon	61.79%	1.0%	−0.46%	−0.62%
	53.2%	50.5%	Carter	48.95%	−53.2%	−1.14%	0.05%
	76.8%	51.6%	Reagan	55.30%	23.2%	1.73%	1.01%
	83.2%	59.0%	Reagan	59.17%	16.8%	0.31%	−0.25%
Bush	81.1%	56.0%	G.H.W. Bush	53.90%	18.9%	−0.16%	−0.77%
Bush	7.8%	43.0%	Clinton	46.54%	−7.8%	−1.02%	−0.97%
	7.0%	44.1%	Clinton	45.34%	−7.0%	2.27%	2.30%
…ush	61.5%	51.1%	Gore	49.73%	38.5%	−0.55%	−1.65%
…ush	55.0%	50.0%	G.W. Bush	51.24%	45.0%	1.15%	−0.11%

(a) Appendix table, part 1



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Appendix
(CONTINUED)

Gallup poll: expected GOP vote share (of the 2 party vote)	Winner	GOP popular vote share (2 party vote share)	Partisan shock I(GOP Pres) − p(GOP Pres)	percentΔStock (−I to t + I)	Regression residual (Table 5, col. 3)
46.2%	n = 14	45.2%	−32.8%	−0.71%	−0.03%
(1.7)		(1.1)	(6.6)	(0.57)	(0.51)
51.6%	n =5	48.6%	−59.5%	−1.44%	−0.09%
(1.1)		(0.4)	(7.4)	(0.82)	(0.65)
44.7%	n = 9	43.3%	−18.0%	−0.31%	0.00%
(1.7)		(1.4)	(4.4)	(0.75)	(0.74)
54.5%	n = 18	55.7%	23.7%	+0.76%	0.03%
(1.5)		(1.17)		(0.27)	(0.30)
n.a.	n = 0	n.a.	n.a.	n.a.	n.a.
54.5%	n = 18	55.67%	23.7%	+0.76%	0.03%
(1.5)		(1.17)	(3.5)	(0.27)	(0.30)

…e analyze data provided to us by Rhode and Strumpf who collected press reports of the Curb Market on Wall Street. …market: For 1976–1988, we rely on press reports of betting odds with British bookmakers; for 1992–1996, we use …ctions of the winner of the popular vote). In 2000 we use data provided by Centrebet, an online bookmaker and for …ion market data for the 1964–1972; our probability assessments for these periods are derived by estimating the …party predicted vote shares from the final pre-election Gallup poll.
…e inverse normal cdf. Using nonlinear least squares, we estimated σ = 4.9, with a standard error of 1.0. Note that …ts, since in 1964 and 1972 the eventual winner was at least 20 points ahead in front in the final Gallup poll, while

…he day prior to the election to the close the following day. (For most of the sample, financial markets were closed …dex come from CRSP for 1964–2004 and Schwert [1990] for 1888–1960, and for 1880–1884, we use Kalinke's [2004]

…on's final pre-poll projections of the two-party vote share. Historical data are from www.gallup.com.
…rned by the two most popular candidates. For the 1912 election, Taft, the Republican candidate was effectively the …publican candidate.
…Republican president from the day prior to the election to the day after. We measure this shock as I(Republican …probability of a Republican president comes from political prediction markets.

(b) Appendix table, part 2

9 Short summary

The paper uses election-night prediction-market shocks as a clean source of variation in expected political outcomes. This design shows that markets expected Bush/Republican victories to raise equity prices, real bond yields, short-run oil prices, and the dollar relative to Democratic alternatives. The historical evidence since 1880 reinforces the equity-market result: the market response depends on the size of the partisan surprise, not simply on which party wins.

10 Conclusion

10.1 Core conclusion

U.S. presidential elections are not neutral events for financial markets. Prediction-market and close-election evidence indicates that the identity of the winner changes expected financial payoffs, especially in equity markets.

10.2 Incremental contribution revisited

The paper's main advance is not only showing that Republican and Democratic presidents are priced differently, but showing this with a design that avoids the standard reverse-causality problem. By using intraday changes in election probabilities during moments when election news dominates other news, the paper turns prediction markets into an identification device.

10.3 Implications

For political economy, the paper provides evidence against strong policy convergence. For empirical finance, it demonstrates how asset prices can reveal market expectations about policy. For applied empirical work, it highlights the value of combining prediction markets with high-frequency event studies.

10.4 Takeaway

The key lesson is that the causal effect of politics on markets is best estimated from unexpected political news. When the surprise component of elections is measured carefully, financial markets appear to systematically prefer Republican presidents in equity valuations, while the bond-market response becomes especially important after 1980.